Chapter 13 — Recommender Systems and Embeddings

When Netflix fills a row with shows, YouTube picks the next video or Amazon shows "customers who bought this also bought", a **recommender system** predicts which items a user most likely wants from a catalogue too large to browse. This chapter goes from a ratings matrix and a similarity formula, through matrix factorisation and learned **embeddings**, to the production **two-tower + ranking funnel** — and how to evaluate it honestly.

What you'll learn: framing and feedback types; similarity measures; content-based vs collaborative filtering; matrix factorisation (SGD, ALS, implicit ALS — implemented); embeddings and item2vec (implemented); two-tower retrieval and its serving/versioning; the funnel with MMR re-ranking; ANN search (IVF implemented, HNSW, PQ); cold start, popularity bias, feedback loops; offline metrics and A/B tests — and how Netflix, YouTube, Spotify and Amazon do it.

All code is in `code/` (numpy, scikit-learn, PyTorch; no downloads — `code/data.py` generates a small dataset shaped like **MovieLens**, the classic GroupLens movie-ratings set). Outputs are real.

13.1 The recommendation problem

What it is

Users U, items I, and a log of **interactions** (ratings, clicks, plays, purchases). For a user u, produce a short **ranked list** of unseen items u is likely to value.

Why it matters

- **Choice overload**: millions of items, ~ten fit on a phone screen.
- **Business value**: Gomez-Uribe & Hunt (2015) wrote that Netflix's recommender influences the choice behind about 80% of hours streamed, most of the rest coming from search.
- **Two-sided value**: helps creators and sellers in the **long tail** (the many niche items) find an audience.

How to frame it

Framing	Question	Typical loss / metric	Example
Rating prediction (regression)	"How many stars would u give i?"	Squared error / RMSE	The Netflix Prize (2006–2009)
Top-K ranking (the real product)	"Which K items should u see, in what order?"	Ranking losses / precision@K, recall@K, NDCG (§13.11)	Home feeds, "Up next", playlists

Modern systems optimise **ranking**: users see only the top, and most feedback is implicit. Framing questions an interviewer expects:

1. **What is a good outcome?** Clicks are cheap and reward clickbait; watch time, purchase or "save" is closer to value.
2. **What signals exist?** Ratings, views, **dwell time**, skips, purchases, searches, item **metadata** (genre, author, price), user profile, context (device, time, location).
3. **Scale and latency** (a home page must be ready well under a second — System Design Fundamentals). 1,000 items can be scored exhaustively; 100 million cannot — hence the funnel (§13.8).
4. **Freshness** (news stales in hours, movies last years) and **constraints** (diversity, creator fairness, safety, "don't show what they already bought").

The user–item matrix

Rows = users, columns = items; cell (u, i) holds the interaction or is empty.

```
          Movie1  Movie2  Movie3  Movie4
Ana          5       4       .       1
Ben          4       5       1       .
Cy           1       .       5       4
```

It is **huge** (10^9 users × 10^8 items) and **extremely sparse** (ours is ~9% filled; real ones often < 0.1%). Recommendation = **filling in missing cells** well enough that each row's top-scoring empty cells are items the user would enjoy.

`code/data.py`: 600 users, 400 movies, 6 genres. Users have hidden genre tastes; movies have 1–2 genres and hidden quality; popular movies are seen far more (a long tail).

```
affinity = taste @ item_genres.T / item_genres.sum(1) + quality     # hidden "true" liking
stars = clip(round(3 + 1.2 * affinity + noise), 1, 5)                # what a rating would be
p_seen = popularity[None, :] * exp(1.5 * affinity)                   # people watch popular + liked
observed = rng.random(p_seen.shape) < p_seen                          # only these cells are known
```

```
users=600 items=400 ratings=21216
density = 8.840%
top 10% of items receive 38% of all ratings
star histogram: {1: 47, 2: 624, 3: 3054, 4: 6335, 5: 11156}
```

Stars skew high because people mostly watch what they expect to like — **selection bias** (data **missing not at random**): observed ratings aren't a random sample of opinions, in real data too.

TL;DR

Fill in a huge, sparse user–item matrix; show each user the top-scoring empty cells. Frame as **top-K ranking** on a well-chosen outcome; state scale, latency, freshness and constraints first.

13.2 Explicit vs implicit feedback

What it is

- **Explicit**: the user says so — stars, thumbs, reviews.
- **Implicit**: inferred from behaviour — clicks, plays, watch time, purchases, add-to-cart, skips, dwell.

Why it matters

Implicit data is **plentiful**, explicit **scarce**. Netflix replaced stars with thumbs in 2017, saying far more members would give a thumb. Most production recommenders learn mainly from implicit signals — which changes the maths.

	Explicit	Implicit
Volume	Low	Very high
Negative signal	Yes (1 star = dislike)	Mostly **no**: "not watched" may mean "never saw it"
Noise	Rating scale differs by person	Accidental clicks, background plays
Strength	Clear preference	Graded: watched 90% vs 5%, played 30 times
Modelling	Regress on observed cells only	All cells: observed = positive, unobserved = weak negative, with **confidence** weights

How it works — behaviour → training data

1. Define the **positive event** (watched > 50%, purchased, saved).

2. Optionally map intensity to **confidence** c = 1 + α·(play count).
3. **Sample negatives**: random unobserved items, or — much stronger — **impressions without clicks**.
4. Correct **position bias** (top slots get clicks for being on top): log position, or randomise a small traffic slice.

Common mistakes

- Treating every unobserved cell as a full-weight dislike.
- Optimising clicks when the goal is satisfaction.
- Ignoring the negative implicit signals you have: early skips, "not interested", short dwell.

TL;DR

Explicit = scarce, real negatives. Implicit = abundant, positives only, unobserved ≠ negative — use confidence weights and sampled negatives.

13.3 Similarity measures

Measure	Formula	Range	Use when
Dot product	$a \cdot b = \Sigma\, a_k b_k$	$(-\infty, \infty)$	Vector length carries meaning (e.g. popularity)
Cosine	$a \cdot b \,/\, (\|a\|\|b\|)$	$[-1, 1]$	Direction matters, length doesn't (text, rating vectors)
Adjusted cosine / Pearson	cosine after subtracting each user's mean	$[-1, 1]$	Ratings, where some users rate everything high
Jaccard	$\|A \cap B\| \,/\, \|A \cup B\|$	$[0, 1]$	Sets / binary data ("bought", "followed")
Euclidean distance	$\|a - b\|$ (smaller = closer)	$[0, \infty)$	Dense embeddings; for unit vectors $\|a-b\|^2 = 2 - 2\cdot\cos$, so rankings match cosine

Worked numbers

From the §13.1 matrix (0 = not rated):

- cos(Ana, Ben) = (5·4 + 4·5 + 0·1 + 1·0) / (√42 · √42) = 40/42 = **0.952** — similar.
- cos(Ana, Cy) = (5·1 + 1·4) / 42 = **0.214** — different.
- Jaccard on movies *liked* (≥ 4): Ana {1, 2} vs Ben {1, 2} → **1.0**; Ana vs Cy {3, 4} → **0**.

Why subtract the mean? A generous rater gives 4–5 to everything, a harsh one 2–3; raw cosine sees them as different even if they order movies identically. E.g. [5, 4, 3] vs [3, 2, 1]: raw cosine ≈ 0.98; mean-centred both become [1, 0, −1] → **1.0**.

Dot product vs cosine: dot product keeps length, so a broadly popular item with a long vector scores higher for everyone — a signal or a bias, depending on the goal.

Common mistakes

- Raw cosine where 0 means "missing" — zeros drag similarity down; mean-centre on observed entries only.
- Trusting similarities from 2 co-rated items — **shrink** toward 0 when overlap is small, or require minimum overlap.
- Training with dot product but serving with cosine (or vice versa).

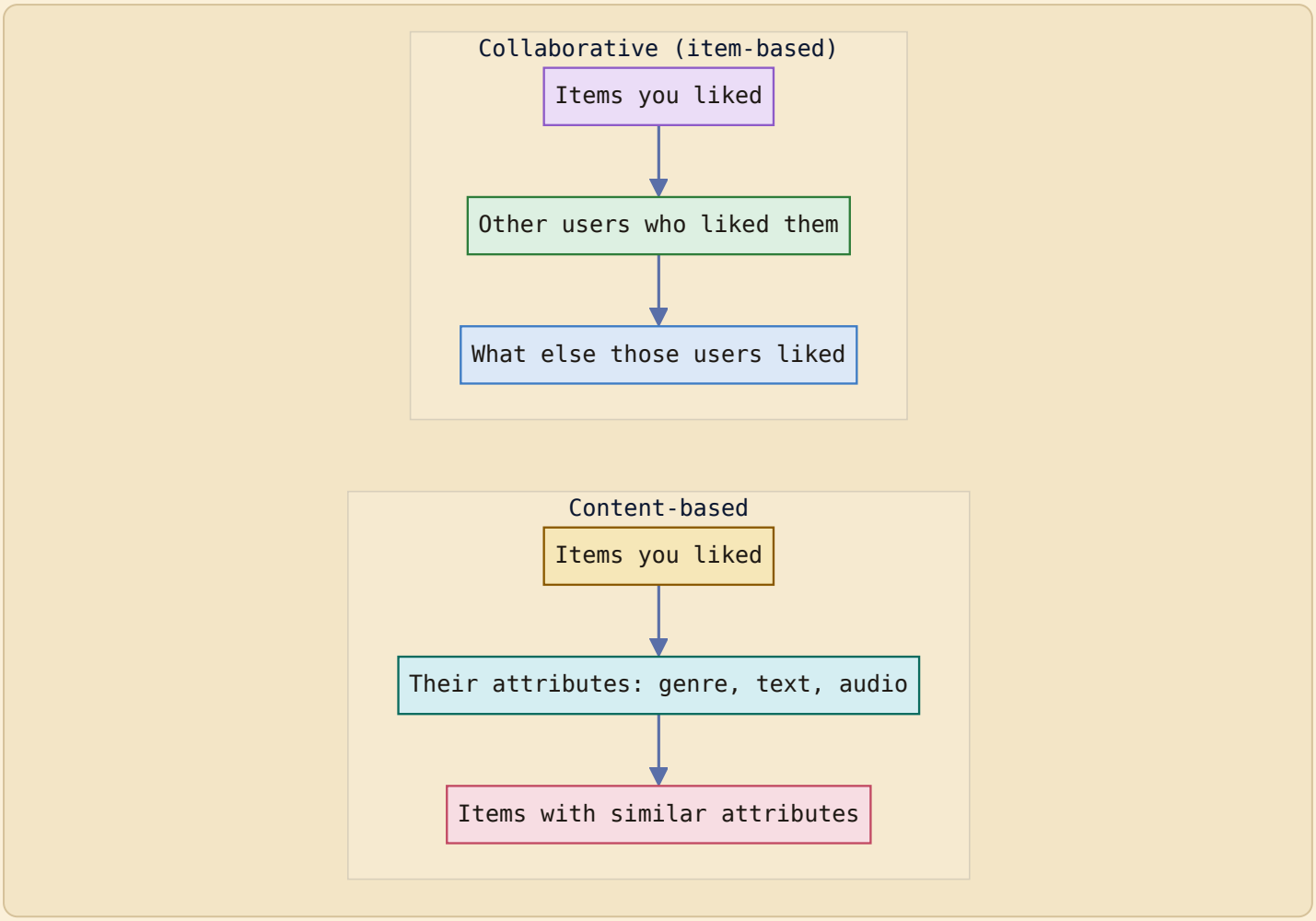
TL;DR

Cosine for direction, dot for magnitude, adjusted cosine for ratings, Jaccard for sets. Mean-centre and shrink small-overlap similarities.

13.4 Content-based vs collaborative filtering

What it is

- **Content-based filtering**: recommend items with **similar attributes** to what the user liked (genre, actors, text, audio, image).
- **Collaborative filtering (CF)**: ignore attributes, use **other users' behaviour** — "people who liked what you liked also liked X".
  - **User-based CF**: find similar users, recommend what they liked.
  - **Item-based CF**: for each item u liked, find items the same users liked. Made famous by Amazon.
- **Hybrid**: blend both, or feed both signals into one model.



How item-based CF predicts a rating

1. Item–item similarity S(i, j) (adjusted cosine over users who rated both).
2. Take the k items most similar to i **that u rated**.
3. $\hat{r}(u, i) = \text{mean}(u) + \Sigma_j S(i, j) \cdot (r(u, j) - \text{mean}(u)) / \Sigma_j |S(i, j)|$ (absolute value keeps the denominator positive; our code keeps only positive neighbours).

Item-based usually beats user-based: items have more ratings each, item similarities are more stable, and they can be precomputed **offline** (batch job) rather than **online** (while the user waits). User–user similarity over 100M users is $O(U^2)$.

Code — `code/similarity_cf.py`

```
# collaborative: item-item adjusted cosine on mean-centred ratings
Rc = np.where(mask, R - user_mean[:, None], 0.0)
norms = np.linalg.norm(Rc, axis=0) + 1e-12
S_cf = (Rc.T @ Rc) / np.outer(norms, norms)
# content-based: item-item cosine on genre vectors
Gn = G / np.linalg.norm(G, axis=1, keepdims=True)
S_cb = Gn @ Gn.T
# ... predict each held-out rating with the k=20 most similar items the user rated
```

```
== Rating prediction on held-out ratings (RMSE, lower is better) ==
predict global mean      : 0.833
predict user's mean      : 0.811
content-based item-kNN   : 0.6872
collaborative item-kNN   : 0.6820
hybrid (average of both) : 0.6711

Movie 51 (Drama) - 5 nearest neighbours
  collaborative: ['40:Drama', '341:Drama', '321:Drama', '325:Drama', '270:Drama']
  content-based: ['109:Drama', '40:Drama', '52:Drama', '56:Drama', '341:Drama']
```

- Both beat the baselines clearly ("item-kNN" = predict from the k nearest items); the **hybrid** beats both — they capture partly different information.
- **CF discovered "Drama" by itself** from who-rated-what, never seeing a genre label.
- Here genre drives taste, so content does well. On real data CF usually wins for items with history (behaviour captures quality and mood metadata misses); content wins for new items.

Trade-offs

	Content-based	**Collaborative filtering**
Needs	Item attributes	Many user interactions
New item (no history)	✅ Works immediately	❌ Cold start
New user	⚠️ Needs a few likes	❌ Cold start
Serendipity (pleasant surprise)	Low — "more of the same" (**filter bubble**)	High — cross-genre finds
Captures quality/taste beyond metadata	❌	✅
Explainability	"Because it's sci-fi like X"	"People like you also liked X"
Popularity bias	Low	High (popular items co-occur with everything)

When not to use it

- Pure CF when the catalogue turns over daily (news, listings) — most items are always cold.
- Pure content when metadata is poor or quality varies wildly within a genre.

Common mistakes

- Re-recommending already-consumed items.
- User–user similarity at huge scale — go item-based, or embeddings + ANN.
- A random split for a "predict what they do next" product — use a **time-based** split (§13.11).

Real-world examples

- **Amazon** "Customers who bought this also bought": item-to-item CF (Linden, Smith & York, *IEEE Internet Computing*, 2003) — item–item similarities precomputed offline so the online step is a cheap lookup.

- **Pandora**'s Music Genome Project: content-based, analysts tagging songs on hundreds of attributes.
- Most modern systems are **hybrids**.

In the interview

> **30-second answer:** "Content-based matches item attributes to what the user liked — handles new items but over-specialises. Collaborative filtering uses co-behaviour — item-based CF is the scalable classic — and finds surprising, quality-aware matches but has cold start and popularity bias. Production combines both, often in an embedding model whose item side uses ID and content features."

- *Why item-based over user-based?* More interactions per item, more stable, precomputable offline.
- *Scaling CF?* Precompute top-N neighbours per item, or embeddings + ANN index (§13.9).
- *Rich metadata, sparse interactions?* Start content-heavy; blend in CF as data grows.

TL;DR

Content = similar attributes; collaborative = similar people's behaviour. CF finds structure (even genres) from behaviour but needs history; content covers cold items. Hybrids win.

13.5 Matrix factorisation

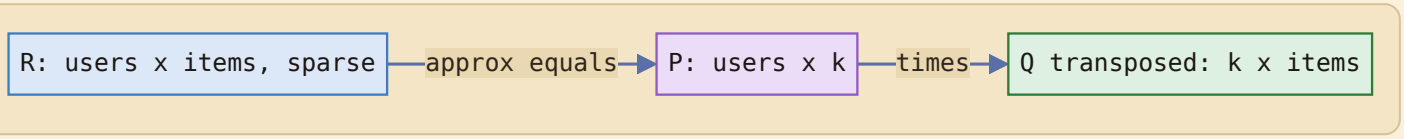
What it is

Matrix factorisation (MF) describes each user and item by a short vector of k hidden **latent factors** (k ≈ 10–200):

$\hat{r}(u, i) = \mu + b_u + b_i + p_u \cdot q_i$

- μ = global mean; b_u = user bias (a generous rater); b_i = item bias (a universally loved film).
- p_u, $q_i \in \mathbb{R}^k$ = user and item factor vectors.

So R (users × items) ≈ **P (users × k) · Qᵀ (k × items)**. Each person gets "taste coordinates" and each item "style coordinates" on a few hidden axes that training discovers.



Why it matters

- Star of the **Netflix Prize** (Simon Funk's 2006 SGD "SVD" post; Koren, Bell & Volinsky, 2009).
- **Generalises** CF: two users with no co-rated movie can still be similar if their vectors align.
- p_u, q_i are **embeddings** — the bridge to two-towers (§13.7); storage drops from U·I to (U + I)·k.

How it works — the objective

Squared error on **observed** ratings only, plus L2 regularisation:

$L = \Sigma_{(u,i)\ \text{observed}} [\ (r - \hat{r})^2 + \lambda(\|p_u\|^2 + \|q_i\|^2 + b_u^2 + b_i^2)\ ]$

(Penalty inside the sum as in Koren et al. 2009 — what the per-rating SGD updates minimise; the factor 2 from differentiating is absorbed into the learning rate η.)

This is **not classical SVD**: SVD needs a full matrix, and zero-filling tells the model "everyone hates everything unseen". Two optimisers:

1. SGD — one rating at a time:

1. $err = r - \hat{r}(u, i)$
2. $b_u \leftarrow b_u + \eta(err - \lambda b_u)$; $b_i \leftarrow b_i + \eta(err - \lambda b_i)$
3. $p_u \leftarrow p_u + \eta(err \cdot q_i - \lambda p_u)$; $q_i \leftarrow q_i + \eta(err \cdot p_u - \lambda q_i)$ (use the old p_u)

2. Alternating least squares (ALS) — the objective isn't **convex** in P and Q jointly, but with one frozen it's a **ridge regression** with a closed-form solution:

1. Freeze Q. For each user: $p_u = (Q_u^T Q_u + \lambda I)^{-1} Q_u^T r_u$ (Q_u = rows of Q for items u rated; r_u = u's ratings on them, minus μ in our bias-free version).
2. Freeze P; solve each q_i the same way.
3. Repeat until error stops falling (10–20 rounds).

Each solve is independent, so ALS **parallelises** trivially (Spark MLlib ships it).

	SGD	ALS
Update	One rating at a time	Whole vector solved exactly
Parallelism	Harder (conflicting updates)	Embarrassingly parallel
Tuning	Learning rate, epochs	Few knobs, few iterations
Implicit data (all cells)	Needs negative sampling	Efficient (Hu et al. 2008)
Extra features, custom losses	Flexible	Less flexible

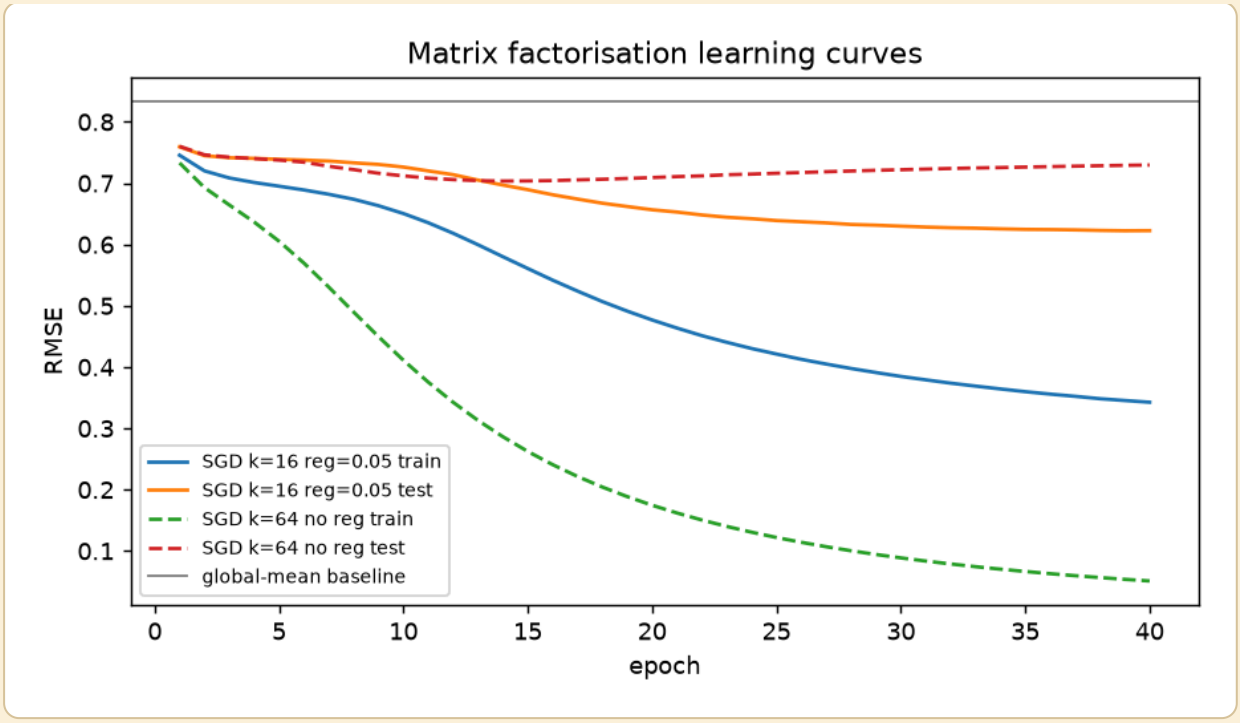
Code — `code/matrix_factorization.py`

```
def mf_sgd(train, test, n_users, n_items, k=16, lr=0.02, reg=0.05, epochs=40, seed=0):
    ...
    for _ in range(epochs):
        for idx in rng.permutation(len(tr)):          # one rating at a time
            u, i, r = tu[idx], ti[idx], tr[idx]
            err = r - (mu + bu[u] + bi[i] + P[u] @ Q[i])
            bu[u] += lr * (err - reg * bu[u])
            bi[i] += lr * (err - reg * bi[i])
            pu = P[u].copy()
            P[u] += lr * (err * Q[i] - reg * P[u])
            Q[i] += lr * (err * pu - reg * Q[i])

def mf_als(train, test, n_users, n_items, k=16, reg=5.0, iters=10, seed=0):
    ...
        for u, idx in enumerate(by_user):               # fix Q, ridge-solve each user
            Qi = Q[ti[idx]]
            P[u] = np.linalg.solve(Qi.T @ Qi + I_k, Qi.T @ (tr[idx] - mu))
        # ... then the same loop over items with P fixed
```

```
baseline (global mean) test RMSE: 0.833
SGD epoch  1: train RMSE 0.746  test RMSE 0.759
SGD epoch  5: train RMSE 0.695  test RMSE 0.739
SGD epoch 10: train RMSE 0.650  test RMSE 0.726
SGD epoch 20: train RMSE 0.477  test RMSE 0.657
SGD epoch 40: train RMSE 0.343  test RMSE 0.623
SGD k=64, no regularisation, epoch 40: train 0.051  test 0.730
ALS iter   1: train RMSE 0.769  test RMSE 0.818
ALS iter   2: train RMSE 0.576  test RMSE 0.724
ALS iter   5: train RMSE 0.496  test RMSE 0.677
ALS iter  10: train RMSE 0.488  test RMSE 0.675

Top-5 for user 0: [98, 325, 236, 40, 314] predicted [5.16 5.14 5.11 4.98 4.93]
true (hidden) stars of those movies: [5. 5. 5. 5. 4.]
```



- Regularised SGD reaches **0.623** — better than item-kNN (0.682) and the mean (0.833).
- ALS converges in ~5 iterations (exact solves); ends slightly worse because this version has no biases.
- **Overfitting** (dashed lines): 64 factors without regularisation drive train error to 0.05 while test climbs to 0.73. With many parameters and sparse data, λ isn't optional.
- User 0's top-5 truly rate 5, 5, 5, 5, 4.

Implicit-feedback MF (weighted ALS)

Hu, Koren & Volinsky (2008) cover **every** cell:

- preference p_{ui} = 1 if u interacted with i, else 0;
- confidence c_{ui} = 1 + α·(count) — observed cells matter a lot, unobserved a little;
- minimise Σ over **all** (u, i) of $c_{ui}(p_{ui} - x_u \cdot y_i)^2 + \lambda(\ldots)$.

Solved with ALS. Speed trick: $Y^T C_u Y = Y^T Y + Y^T(C_u - I)Y$ — $Y^T Y$ is shared by all users and the second term touches only u's few items. `code/implicit_eval.py` uses the dense form; libraries: `implicit`, Spark MLlib `ALS(implicitPrefs=True)`.

Trade-offs / when not to use it

- MF sees only IDs: cold items have no vector, and context (time, device) is hard to add — use feature-based models (**factorisation machines**, which learn pairwise interactions between any features, and two-towers).
- A dot product is a limited interaction function, yet tuned MF stays a strong baseline: Rendle et al. (2020) found a tuned dot product beat MLP-based "neural collaborative filtering".

Common mistakes

- Zero-filling and running `np.linalg.svd` — teaches "unseen = hated".
- No regularisation, or tuning k on the test set (sweep k on validation; larger k needs more λ).
- Dropping biases — much of the signal is "this user rates high" and "this movie is loved".

Real-world examples

- **Netflix Prize**: *BellKor's Pragmatic Chaos* won in 2009 with the required 10% RMSE improvement over Cinematch, using a large blend of factor and neighbourhood models. Netflix wrote in 2012 that it productionised two components from an earlier progress prize (SVD-style factorisation and restricted Boltzmann machines) but not the final blend — the extra accuracy didn't justify the engineering cost.

- **Spotify** used implicit-feedback MF on listening data as one pillar of Discover Weekly.

In the interview

> **30-second answer:** "MF gives every user and item a k-dimensional vector and predicts affinity as a dot product plus biases, trained on observed entries with L2 — by SGD, or by ALS, which becomes many small parallel ridge regressions. For implicit data, weighted ALS with confidence weights. Weak spots: cold start and no context, which feature-based two-towers fix."

- *Why not plain SVD?* It needs a complete matrix; imputing zeros biases toward "dislike".

TL;DR

Users and items as short vectors; affinity = dot product + biases. Train on observed cells with L2 by SGD (flexible) or ALS (parallel, great for implicit). Strong baseline; fails on cold start.

13.6 Embeddings

What it is

An **embedding** is a short, dense, learned vector (tens to a few hundred numbers) representing a discrete thing — user ID, item ID, word, category. MF's p_u and q_i are embeddings.

The problem it solves

Raw IDs carry no meaning (`item_5521` isn't closer to `item_5522` than to `item_9`). **One-hot** vectors are enormous for 100M items and every pair is equally distant. Embeddings are small and training places them so **distance reflects the training objective**: items serving the same users end up near each other. Mechanically an embedding layer is a lookup table (one row per ID); multiplying a one-hot vector by it selects a row. "Close" means close *for that objective* — co-watched items may still differ in price, language or quality.

Four ways to train them

Route	Idea	Best when
Matrix factorisation	Factor the interaction matrix (§13.5)	Only interactions; strong baseline
Two-tower (contrastive)	Real (user, item) pairs must outscore sampled negatives (§13.7)	Large-scale retrieval with features
Graph embeddings	Random walks or graph neural networks over users, items, follows, co-purchases	Relationships are the signal (Pinterest's PinSage)
Pretrained content encoders	Text/image/audio model embeds item content	Brand-new items; text/image-heavy catalogues

Text embeddings from pretrained language models (semantic search, RAG) are in the AI book (embeddings, vector databases and RAG); the geometry is the same.

word2vec → item2vec

word2vec (Mikolov et al., 2013) skip-gram with negative sampling — centre word predicts window context, σ(centre · context) → 1 for real pairs and → 0 for k frequency^0.75-sampled negatives — is taught in §12.9. **item2vec** (Barkan & Koenigstein, 2016) treats a session, playlist or basket as a "sentence" and its items as "words", so co-consumed items get nearby vectors — no labels needed. Airbnb (listing embeddings from search click sessions, Grbovic & Cheng, KDD 2018) and Spotify (tracks co-occurring in playlists) describe using it; case study 18.9 builds a similar-listings carousel on it.

`code/item2vec.py` implements it in numpy:

```
pos = sig(np.sum(vc * W_out[o], 1))                 # centre . true context  -> want 1
ngs = sig(np.einsum("bd,bkd->bk", vc, W_out[negs]))  # centre . 5 negatives   -> want 0
g_pos = (pos - 1)[:, None]                          # gradient of -log(pos)
g_neg = ngs[:, :, None]                             # gradient of -log(1 - ngs)
grad_c = g_pos * W_out[o] + np.sum(g_neg * W_out[negs], 1)
np.add.at(W_out, o, -lr * g_pos * vc)               # np.add.at handles repeated ids
np.add.at(W_out, negs, -lr * g_neg * vc[:, None, :])
np.add.at(W_in, c, -lr * grad_c)
```

```
epoch  1  mean loss per pair 2.884
epoch 10  mean loss per pair 2.581
top-5 neighbours sharing the main genre: 79% (random would give ~17%)
item 51 (Drama) neighbours: ['113:Drama', '236:Drama', '281:Action', '321:Drama', '146:Drama']
```

Seeing only lists of item IDs, 79% of nearest neighbours share the item's genre (chance 17%) — **meaningful geometry learned from behaviour**.

When not to use embeddings

- **Small catalogues** where exact scoring already fits the budget.
- **Numeric inputs** (price, distance, account age) — already meaningful; scale and use directly.
- **Decisions that must be explained** — embedding dimensions have no human meaning; credit and fraud keep interpretable features alongside.

Common mistakes

- Comparing vectors from two different trainings — each run has its own coordinate system (§13.7).
- An embedding per rare ID (noisy) — use a minimum count, **hashing** (many IDs into a fixed number of shared buckets), or content features.
- Assuming "near" means "similar in every way".

TL;DR

Short learned vectors giving IDs a geometry where near = related (for the objective). Learn via MF, two-tower, graphs or content encoders; item2vec = word2vec on sessions.

13.7 Two-tower retrieval

What it is

A **two-tower model** (*dual encoder*) has two separate networks:

- **user (query) tower**: user features — ID, recent clicks, search terms, device, hour → vector;
- **item (candidate) tower**: item features — ID, category, text/image content → vector of the **same length** (64–256).

Score = **dot product** (or cosine) of the two.

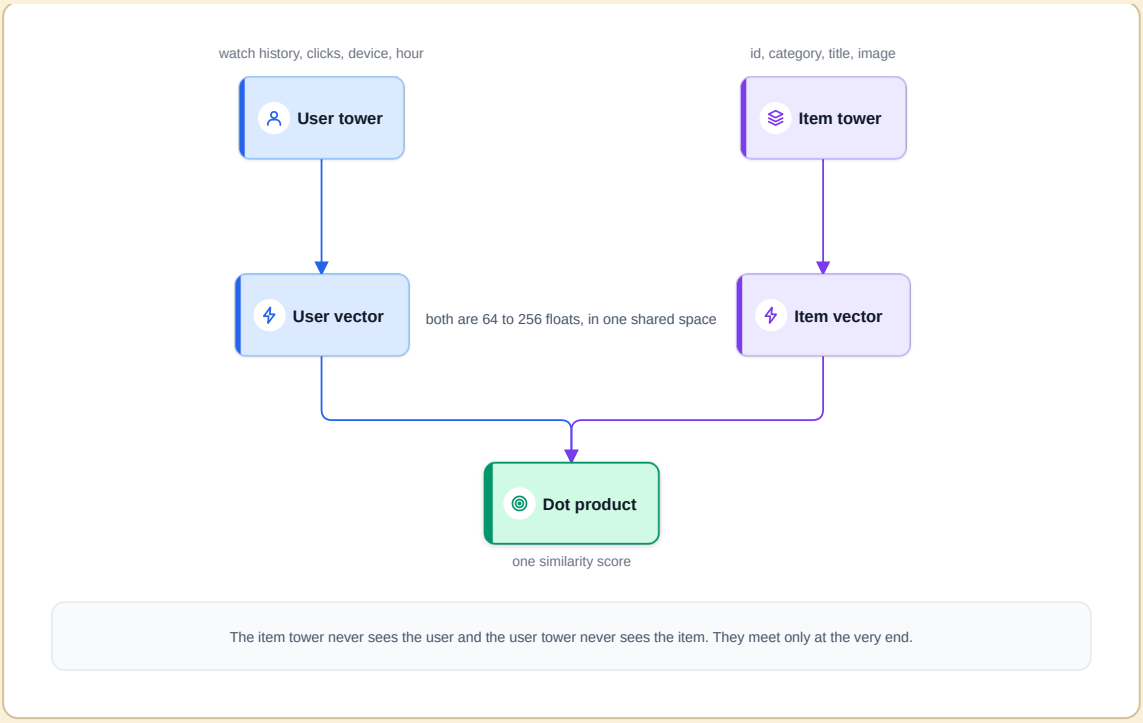


Figure — The towers run independently and meet only at the final dot product (course diagram).

Why it matters — the key property

The towers **never see each other's inputs**. So the item tower embeds the **entire catalogue ahead of time** into a nearest-neighbour index; per request only the user vector is computed and the index returns its nearest items — 100M items searched in milliseconds.

A **cross model** reads user and item features together and learns interactions ("this user × this item's price") — more accurate, but nothing precomputes; it would run per item per request. Hence two-towers do **retrieval**, cross models **ranking** (§13.8).

How it is trained

Logs hold positive pairs (user, video watched), (user, product bought).

1. Take a batch of B positive pairs; encode → B user and B item vectors.
2. Score all users against all items: a B × B dot-product matrix divided by a **temperature** τ (0.05–0.1; small τ sharpens the softmax).
3. Row j's correct "class" is column j; the other B − 1 items are negatives; softmax cross-entropy. This is **in-batch negative sampling** — negatives for free.
4. **logQ correction** (Yi et al., 2019): a hit item appears as a negative in batch after batch and gets pushed down too hard, so subtract log(item's sampling probability) from its logit.
5. Optionally a few **hard negatives** (close but wrong, e.g. shown-not-clicked) — useful, but too many or mislabelled ones hurt. Add them last.

Code — `code/two_tower.py`

```
class Tower(nn.Module):
    def __init__(self, n_ids, n_feats, id_dropout=0.0):
        super().__init__()
        self.id_emb = nn.Embedding(n_ids, 32)
        self.mlp = nn.Sequential(nn.Linear(32 + n_feats, 64), nn.ReLU(), nn.Linear(64, DIM))
        self.id_dropout = id_dropout   # randomly hide the id so the tower also learns from content

    def forward(self, ids, feats, use_id=True):
        e = self.id_emb(ids)
        if not use_id:
            e = torch.zeros_like(e)
        elif self.training and self.id_dropout:
            e = e * (torch.rand(len(ids), 1) > self.id_dropout).float()
        return F.normalize(self.mlp(torch.cat([e, feats], 1)), dim=1)

# training step
uv = user_tower(u, U_feat[u])                # user tower: id + genre profile of history
iv = item_tower(i, G[i])                     # item tower: id + genre multi-hot
logits = uv @ iv.T / temperature             # (B, B) in-batch scores
logits = logits - log_q[i][None, :]          # logQ popularity correction
loss = F.cross_entropy(logits, torch.arange(len(idx)))
```

40 items are held out entirely as brand-new items, scored with the item tower's ID part zeroed (content only).

```
epoch  1  in-batch softmax loss 6.637
epoch  5  in-batch softmax loss 6.022
epoch 15  in-batch softmax loss 5.798

warm items  recall@50: two-tower 0.649 | popularity 0.510
cold items  hit@10 among the 40 new items: 0.586 (random = 0.250)
```

- Known items: recall@50 0.65 vs 0.51 for "most popular".
- **Never-seen items**: content alone puts the right new item in the top 10 of 40 for 59% of users (random 25%) — two-towers soften **item cold start**. `id_dropout` is what makes the content-only vector meaningful.
- (Batch 512 → uniform guessing loss ln 512 ≈ 6.24, but logQ shifts the logits, so read the trend and judge by recall.)

How it serves

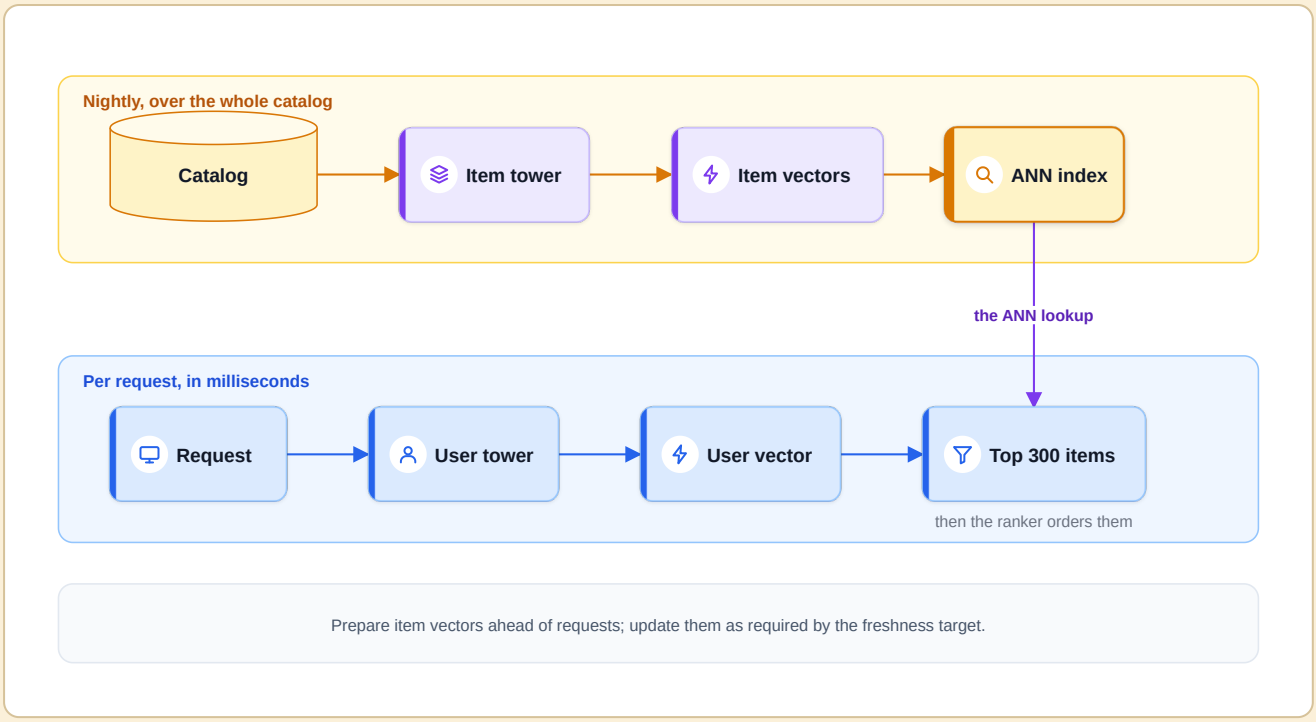


Figure — Offline item embedding and index build vs the online per-request path (course diagram).

1. **Offline (e.g. nightly):** embed the catalogue, rebuild the **ANN index** (approximate nearest neighbour, §13.9).
2. **Per request:** run the user tower (or reuse a recently cached vector).
3. **Lookup:** a few hundred nearest items in milliseconds — the **candidates** for the ranker.

Freshness differs by tower. Item vectors refresh as often as catalogue change and measured retrieval quality demand (nightly is an example). The user side must be fresher — three cooking videos just watched should yield cooking candidates *now* — so the user tower reads **real-time features** from an online **feature store** at request time.

The versioning trap

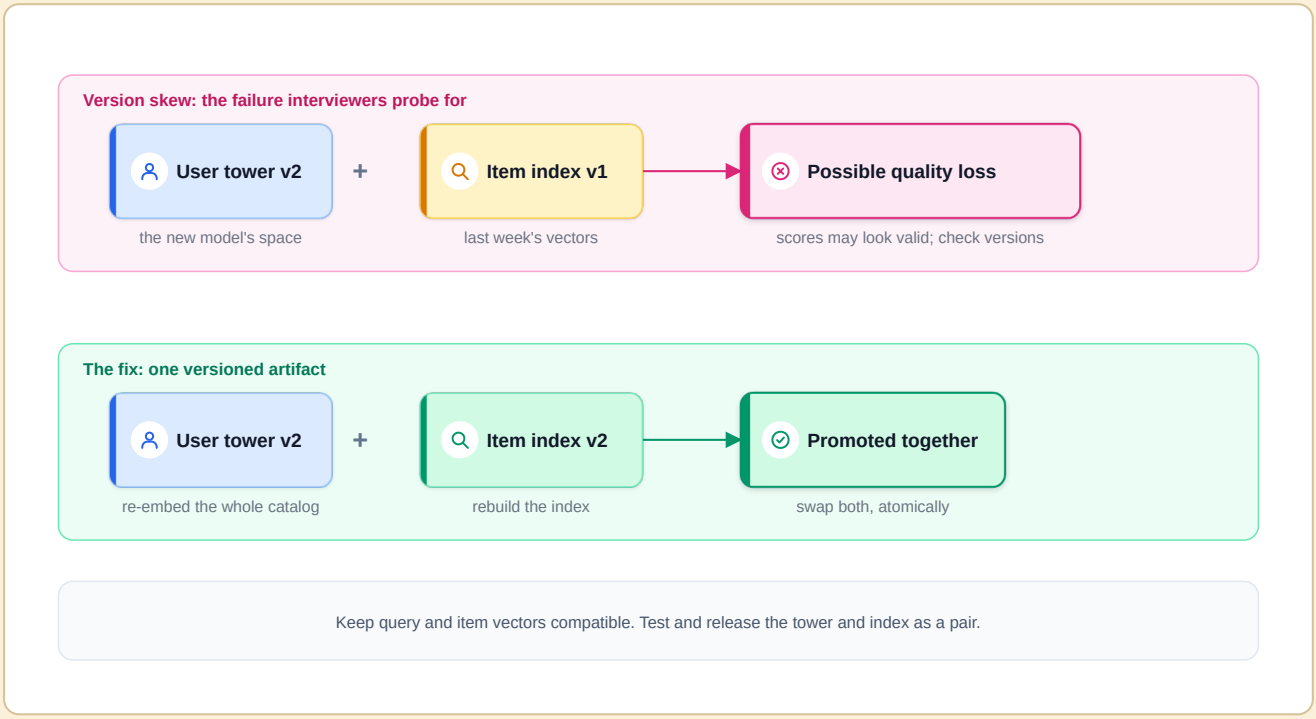


Figure — Version skew: a new user tower querying an index built by an old item tower (course diagram).

An embedding means something only **inside its model's space**; retraining rotates and rescales it. Upgrading is a coordinated migration: (1) re-embed the catalogue with the **new** item tower, (2) build a **new** index, (3) switch new user tower and new index **together, atomically** (never half-switched) as a versioned pair in the **model registry**, keeping the old pair for **rollback**.

Version skew (new tower on old index, or the reverse) is dangerous because nothing crashes — scores are valid numbers, latency is normal, only quality silently drops. Defences: version tags on requests and indexes, routing to compatible pairs, offline validation of the pair, retrieval-quality monitoring during rollout. Budget re-embedding compute (100M items is not free).

Trade-offs

	Two-tower	Cross model (ranker)
User–item feature interactions	Only via final dot product	Full (any feature crosses)
Precompute items	✅	❌
Items scored per request	Whole catalogue (via ANN)	A few hundred
Role	Retrieval / candidate generation	Ranking

When not to use it

Small catalogues (score everything exactly), or when co-occurrence is good enough and training + indexing + versioning machinery isn't affordable.

Common mistakes

- The same item twice in a batch becomes its own false negative — deduplicate or mask.
- No popularity correction → hits pushed down (or, with uniform random negatives, pushed up).
- Mixing tower and index versions; leaking the future into user features (events after the label).

Real-world examples

- **YouTube**: Covington, Adams & Sargin (2016) — candidate generation as a huge softmax over videos, served by nearest-neighbour search, then a ranking network; Yi et al. (2019) — two-tower with streaming logQ.
- **Pinterest, LinkedIn, Airbnb** and many e-commerce sites describe embedding-based retrieval.

In the interview

> **30-second answer** ("How do you find candidates among 100M items?"): "A two-tower model trained with in-batch negatives and logQ correction. Item vectors go offline into an ANN index; per request I compute a fresh user vector and pull a few hundred nearest items for the ranker. I'd watch retrieval recall and refresh cadence, and ship tower and index as a tested, versioned pair."

- *What keeps the sides consistent?* Tower + index as one versioned **artifact**, switched atomically, with a rollback pair.
- *Can retrieval surface a zero-interaction item?* Yes, via content features; add a "new arrivals" slot or exploration traffic.
- *Why not one network over user and item?* Can't precompute — use it as the ranker on the few hundred retrieved.

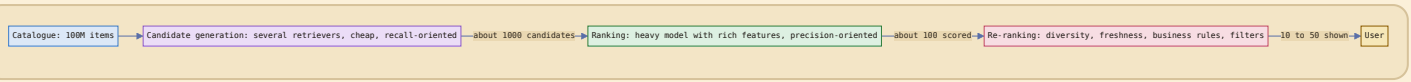
TL;DR

Two towers encode users and items independently; they meet only at a dot product, so items go offline into an ANN index and only the user tower runs per request. In-batch negatives + logQ; release tower and index together.

13.8 The funnel: candidate generation → ranking → re-ranking

What it is

Large recommenders are **multi-stage funnels**; each stage sees fewer items and can afford a heavier model.



Stage	Input → output	Model	Optimises
Candidate generation (retrieval)	$10^8 \rightarrow 10^3$	Several merged sources: two-tower + ANN, item-kNN, **co-visitation** (items viewed in the same sessions), trending, followed creators — each covers a different need, merging raises recall	**Recall**: don't lose good items
Ranking	$10^3 \rightarrow$ ordered 10^2	GBDTs or deep nets (Wide & Deep, DCN, DLRM) on user × item × context features; predicts P(click), watch time, P(purchase)… combined into one score	**Precision** at the top
Re-ranking	$10^2 \rightarrow$ 10–50	Rules + algorithms (MMR, quotas)	Diversity, freshness, fairness, policy, de-duplication

Some systems add a **pre-ranking** ("light ranking") stage when candidate counts are very large. No single model is both cheap enough for 100M items and rich enough for fine feature crosses — the funnel spends compute where it matters. Retrieval sets a **ceiling**: an item not retrieved can never be ranked.

Maximal Marginal Relevance (MMR)

A ranker scores items independently, so the top 10 are often near-duplicates (ten episodes of one show) — diversity therefore belongs in re-ranking. **MMR** (Carbonell & Goldstein, 1998) re-ranks greedily:

next item = argmax over remaining j of $[\ \lambda \cdot \text{relevance}(j) - (1-\lambda) \cdot \max \text{similarity}(j, \text{already chosen})\ ]$

$\lambda = 1$ is pure relevance; smaller λ trades relevance for variety.

Code — `code/funnel_rerank.py`

Stage 1: item-kNN retrieves 100 candidates. Stage 2: a logistic-regression ranker trained on half the users scores them with richer features (kNN score, implicit-ALS score, popularity, genre match, average stars). Stage 3: MMR ($\lambda = 0.7$) on the ranker's top 30.

```
def mmr(cands, relevance, sim, lam=0.7, k=K):
    chosen, pool = [], list(range(len(cands)))
    while len(chosen) < k:
        best = max(pool, key=lambda j: lam * relevance[j]
                   - (1 - lam) * max((sim[cands[j], cands[c]] for c in chosen), default=0))
        chosen.append(best); pool.remove(best)
    return [cands[j] for j in chosen]
```

```
stage 1 candidate recall@100 = 0.787  (ceiling for later stages)
stage 1 only (kNN order)           recall@10 0.375  NDCG@10 0.398  diversity 0.728
stage 2 ranker (logistic regr.)    recall@10 0.393  NDCG@10 0.409  diversity 0.524
stage 3 re-rank (MMR, lambda=0.7)  recall@10 0.375  NDCG@10 0.391  diversity 0.683
ranker weights: {'knn_score': 0.18, 'als_score': 0.63, 'log_popularity': 0.36, 'genre_match': 0.62, 'avg_star':
0.07}
```

- Retrieval keeps 79% of held-out items in 100 candidates — the most any later stage can recover.
- The ranker lifts NDCG (0.398 → 0.409) by combining signals, leaning on ALS score and genre match.
- But it makes lists much **less diverse** (0.728 → 0.524), piling onto the favourite genre. MMR restores diversity (0.683) at a small offline cost; online, users often reward variety — an A/B test decides.

Common mistakes

- Training the ranker only on what the *old* system showed (exposure bias) — log impressions and some randomised exposure.
- Evaluating the ranker but not retrieval recall.
- A single objective (clicks) when the business wants long-term satisfaction — use a **multi-task** ranker (click, watch time, like, "not interested") with tuned weights.

In the interview

> **30-second answer:** "No single model is both cheap enough for the whole catalogue and rich enough for fine feature crosses, so: several high-recall retrievers to ~a thousand candidates, a heavy ranker trained on logged impressions, and a re-ranker for diversity, freshness and policy. I track retrieval recall separately — it caps everything after it."

TL;DR

Cheap high-recall retrieval ($10^8 \to 10^3$) → heavy precise ranking ($\to 10^2$) → re-ranking for diversity/rules ($\to 10$). Retrieval caps ranking; MMR buys variety for a little relevance.

13.9 Approximate nearest neighbour (ANN) search

What it is

k-nearest-neighbour search returns the k vectors most similar to a query q. **Exact** (brute-force) search is O(N·d) per query. **ANN** returns *almost* the same top-k while scanning a small fraction. 10^8 items × 128 dims ≈ $1.3 \cdot 10^{10}$ multiply-adds per query is far too slow at thousands of QPS; ANN trades a little **recall** (fraction of the true top-k found) for 10–1000× speed-ups.

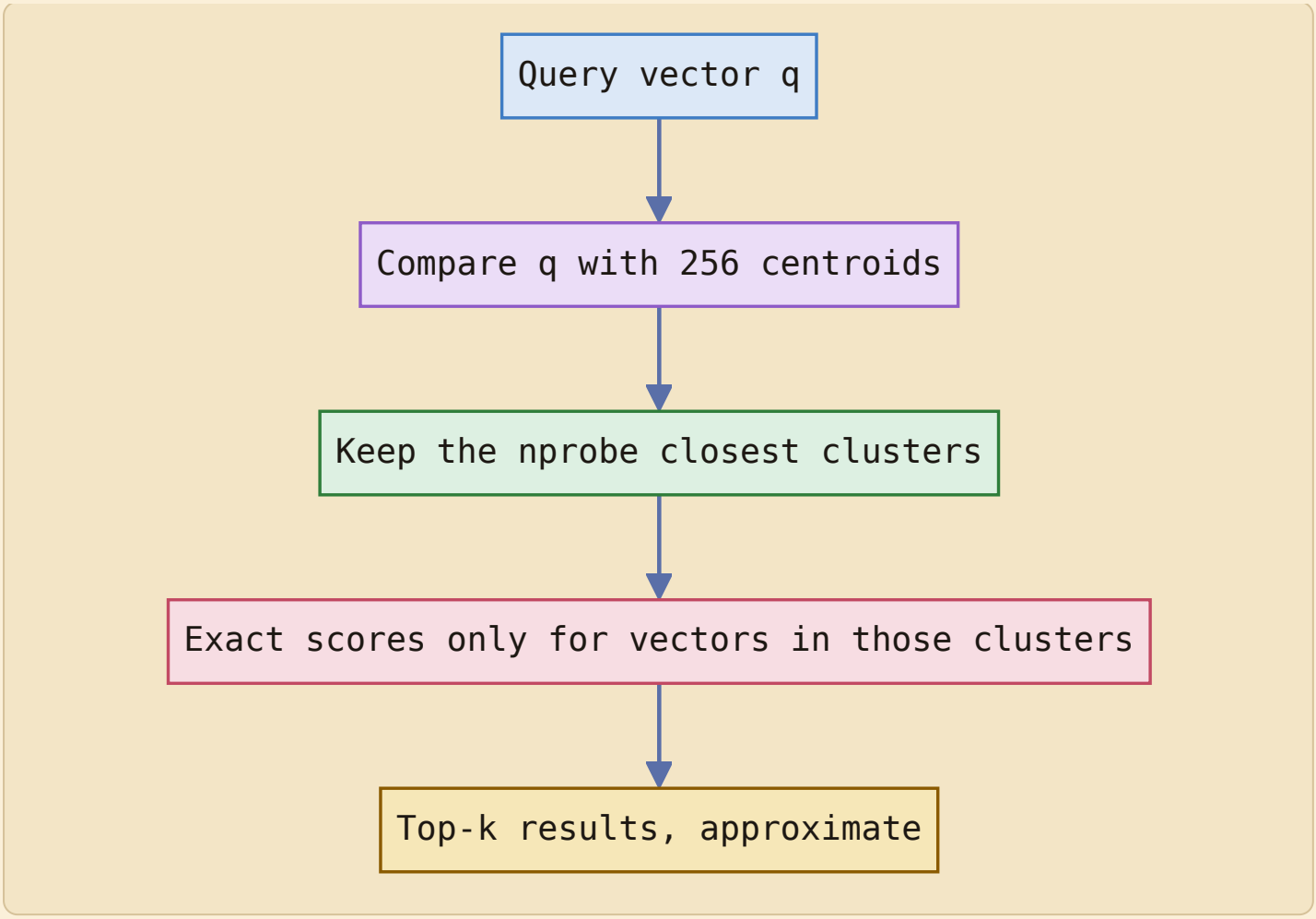
How the main families work

1. IVF — inverted file index (clustering). Offline: **k-means** the vectors into `nlist` centroids; store each vector in its nearest centroid's list. Query: find the `nprobe` nearest centroids, scan only those lists exactly. Larger `nprobe` → higher recall, slower.

2. HNSW — Hierarchical Navigable Small World graphs (Malkov & Yashunin, 2016/2018). Each vector links to ~M close neighbours; upper layers hold few nodes (long-range "express" hops), the bottom holds all. Query: start at the top, greedily hop closer to q, drop a layer when stuck; at the bottom keep a candidate list of size `ef_search`. Knobs M, `ef_construction`, `ef_search`. Excellent recall/latency and incremental inserts; more memory than IVF.

3. Product quantisation (PQ) compresses vectors (128 floats = 512 bytes → 16 bytes) by splitting into sub-vectors, each replaced by its nearest sub-centroid's ID; distances come from small lookup tables. Often combined: **IVF-PQ**.

4. Tree forests (Spotify's Annoy): random **hyperplane** splits build several trees; search a few.

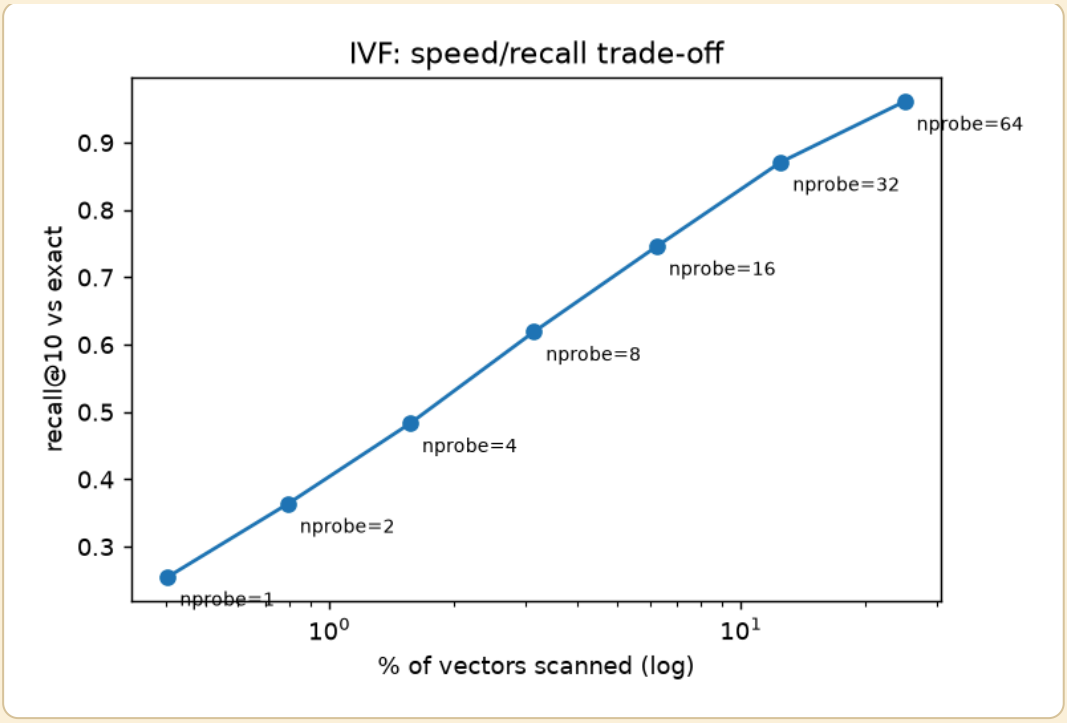


Code — `code/ann_demo.py` (brute force vs IVF)

```
exact = np.argsort(-(queries @ X.T), axis=1)[:, :K]          # brute force: score every vector
C, _ = kmeans(X[rng.choice(N, 20_000, replace=False)], 256)  # train centroids on a sample
assign = np.argmax(X @ C.T, axis=1)
lists = [np.flatnonzero(assign == c) for c in range(256)]    # the "inverted file"
for q in queries:
    probe = np.argsort(-(C @ q))[:nprobe]                    # closest clusters
    cand = np.concatenate([lists[c] for c in probe])
    top = cand[np.argsort(-(X[cand] @ q))[:K]]
```

100,000 unit vectors, 32 dims (timings machine-dependent):

```
brute force: scans 100% of 100,000 vectors, 6.13 ms/query, recall@10 = 1.000
IVF build (k-means, 256 lists): 0.1 s
IVF nprobe= 1: scans  0.41%, 0.06 ms/query, recall@10 = 0.255
IVF nprobe= 2: scans  0.79%, 0.08 ms/query, recall@10 = 0.363
IVF nprobe= 4: scans  1.57%, 0.14 ms/query, recall@10 = 0.483
IVF nprobe= 8: scans  3.14%, 0.27 ms/query, recall@10 = 0.619
IVF nprobe=16: scans  6.25%, 0.56 ms/query, recall@10 = 0.746
IVF nprobe=32: scans 12.49%, 1.21 ms/query, recall@10 = 0.871
IVF nprobe=64: scans 24.96%, 2.56 ms/query, recall@10 = 0.961
```



Scanning 12% gives 87% recall at ~5× speed; 25% gives 96%. (Deliberately hard, overlapping clusters; real embeddings reach high recall at smaller `nprobe`, and production libraries are far faster than this numpy loop.)

Trade-offs

Method	Recall/speed	Memory	Build	Updates
Brute force	Exact, O(N)	Vectors only	None	Trivial
IVF	Good, tune `nprobe`	Low	k-means (fast)	Add to lists; re-train centroids occasionally
IVF-PQ	Good at huge scale	Very low (compressed)	Slower	As IVF
HNSW	Excellent	Higher (graph links)	Slowest	Incremental inserts; deletes awkward

Libraries: **FAISS** (Meta), **ScaNN** (Google), **hnswlib**, **Annoy/Voyager** (Spotify); **vector databases**: Milvus, Weaviate, Qdrant, pgvector, OpenSearch.

When not to use it

Up to ~a million vectors, batched brute-force matrix multiply (especially on GPU) may meet the latency budget — exact and simple. Measure first.

Common mistakes

- A cosine index fed un-normalised vectors (or vice versa).
- Reporting latency without recall@k against exact search.
- Filtering (country, availability) *after* ANN and ending with too few results — use filtered search or over-retrieve.

TL;DR

Exact k-NN is O(N); ANN (IVF clustering, HNSW graphs, PQ compression) scans a fraction for a small recall loss. Always report recall with latency.

13.10 Cold start

Cold start = recommending with little or no history: **new user**, **new item** (CF/MF have no vector), **new system**.

Case	Strategies
New user	Popular/trending by region or device; short onboarding ("pick 3 genres"); context (location, time, referrer); fast updates from first clicks (session-based models)
New item	Content features in the item tower (59% vs 25% random in `two_tower.py`); creator/brand signals; **exploration** — a small controlled traffic share and a "new arrivals" slot
New system	Rules + popularity + content similarity, then collect data

Exploration is essential: if only proven items are shown, new ones never get to prove themselves. **Multi-armed bandits** (ε-greedy, UCB, Thompson sampling — Chapter 14) balance exploiting proven items with exploring newcomers.

Common mistakes

Letting cold items sink forever because their score rests on zero data; long onboarding questionnaires few finish.

TL;DR

New users: popularity + onboarding + context + fast session updates. New items: content features + controlled exploration. "No data" must never mean "never shown".

13.11 Evaluation

What it is

Offline (held-out history — fast, cheap) then **online** (A/B tests — the final judge).

Offline accuracy metrics (top-K)

For one user with ranked `recs` and held-out `relevant` items:

Metric	Formula	Meaning
Precision@k	(# relevant in top k) / k	Quality of what we showed
Recall@k	(# relevant in top k) / (# relevant)	How much of the good stuff we found
Hit rate@k	1 if any relevant in top k	At least one right? (common with leave-one-out)
AP@k (average precision)	(Σ precision@i over ranks i holding a hit) / min(#relevant, k)	Rewards early hits
MAP@k	mean AP@k over users	
NDCG@k	DCG / ideal DCG, DCG = Σ_i rel_i / $\log_2(i + 1)$ (variant gain: $2^{rel_i} - 1$)	Position-discounted, normalised to [0, 1]; supports graded relevance
MRR	mean of 1 / (rank of first hit)	Only the first hit matters (search-like)
RMSE / MAE (root mean squared / mean absolute error)	error on predicted ratings	Rating-prediction framing only

Worked example (`implicit_eval.py`): recs = A B C D E, relevant = {A, C, F}, k = 5.

- Hits at ranks 1 and 3. Precision@5 = **0.400**; Recall@5 = **0.667**; Hit@5 = **1**.
- AP@5 = (1/1 + 2/3) / 3 = **0.556**.
- DCG = $1/\log_2 2 + 1/\log_2 4 = 1.5$; ideal = 1 + 0.631 + 0.5 = 2.131; NDCG@5 = **0.704**.
- (Single hit at rank 4: NDCG@5 = $1/\log_2 5 \approx 0.431$; MRR = 0.25.)

```python
def average_precision_at_k(recs, relevant, k=K):
    hits, total = 0, 0.0
    for rank, item in enumerate(recs[:k], start=1):
        if item in relevant:
            hits += 1
            total += hits / rank          # precision at each position where we hit
    return total / min(len(relevant), k)

def ndcg_at_k(recs, relevant, k=K):
    dcg = sum(1 / np.log2(rank + 1) for rank, it in enumerate(recs[:k], 1) if it in relevant)
    ideal = sum(1 / np.log2(rank + 1) for rank in range(1, min(len(relevant), k) + 1))
    return dcg / ideal
```

Beyond-accuracy metrics

- **Catalogue coverage**: fraction of items shown to *anyone*.
- **Intra-list diversity**: 1 − mean pairwise similarity within a list (here genre cosine).
- **Novelty**: mean popularity of recommendations (lower = more novel); **serendipity** = relevant *and* unexpected.
- **Fairness / exposure**: how impressions spread across creators or sellers.

Code — three models on implicit data (`code/implicit_eval.py`)

Interactions binarised to "watched"; 5 random watched items per user hidden as **test**, ranked among all unseen items. ALS settings chosen on a separate **validation** split (5 more items per user), never on test.

```
ALS chosen on validation split: alpha=10, lambda=50 (validation NDCG@10 0.331)

model              P@10   R@10   MAP@10 NDCG@10 HR@10  cover  divers  avgPop
Popularity         0.107  0.215  0.104  0.197  0.708  0.07   0.741   219
Item-kNN (cosine)  0.184  0.367  0.256  0.388  0.882  0.34   0.733   186
Implicit ALS       0.195  0.390  0.282  0.416  0.893  0.42   0.591   154
(average item popularity in the catalogue = 46 interactions)
```

- **Popularity** is a strong baseline (hit rate 71%) — always include it — but covers 7% of the catalogue with items ~5× more popular than average.
- **Implicit ALS** wins every accuracy metric, covers 42%, recommends less popular items — but lists are less genre-diverse. Accuracy, coverage, novelty and diversity pull apart; choose deliberately.
- The sweep ($\alpha \in \{5, 10, 20\}$, $\lambda \in \{10, 50, 100\}$, $k = 16$) picked $\alpha = 10$, $\lambda = 50$; refit on all training data, scored once on test.

Offline evaluation pitfalls

- **Random splits leak the future** — use a **time-based split**.
- **Sampled metrics** (true item vs 100 random negatives) inflate scores and can flip which model wins (Krichene & Rendle, 2020) — rank against the full catalogue when possible.
- **Logs are biased by the old system**. **Inverse propensity scoring (IPS)** reweights each example by 1 / P(old system showed it) — unbiased, but higher variance.

Online evaluation

- **A/B test**: random user split; business metrics (watch time, **retention**, purchases) plus **guardrails** (latency, complaints, diversity); run long enough for **novelty effects** to fade.
- **Interleaving**: merge two rankers' lists and see whose items get clicked — needs far less traffic; Netflix uses it to pre-screen ideas.

- Offline metrics are a filter, not the verdict — logs reflect the old policy and ignore presentation and long-term effects.

In the interview

> **30-second answer:** "Offline: time-based split, recall@k for retrieval and NDCG@k or MAP@k for ranking over the full catalogue, plus coverage, diversity and novelty, versus a popularity baseline. Then an A/B test on the real objective, such as retention, with latency and diversity guardrails."

- *Precision@k vs recall@k?* Quality of what's shown vs how much of the relevant set was found; retrieval cares about recall, the final list about precision/NDCG.
- *Why NDCG over precision?* Rewards higher placement; supports graded relevance.
- *Model A: higher NDCG, 5% coverage; B: slightly lower NDCG, 40% coverage?* Depends on goals; usually test B online — low coverage starves the long tail and feeds the popularity loop.

TL;DR

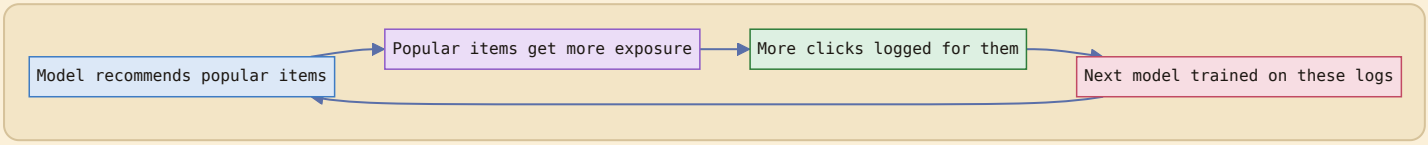
Offline: precision/recall/hit@k, MAP, NDCG on a time-based split over the full catalogue, plus coverage/diversity/novelty and a popularity baseline. Online: A/B tests or interleaving on the true objective.

13.12 Popularity bias and feedback loops

What it is

Popularity bias: models over-recommend popular items — from the data (more interactions → more updates and co-occurrences) and the model (dot products favour long vectors).

Feedback loop: popular items shown → more clicks → next model rates them higher → shown more. Long-tail items never collect data to prove themselves → less diversity, filter bubbles, struggling niche creators.



How to counter it

1. **Measure**: average popularity of recommendations, coverage, Gini coefficient of exposure (0 = even, 1 = all to one item).
2. **Correct in training**: logQ (§13.7), down-weighting popular positives, IPS.
3. **Explore**: a little bandit traffic for new and long-tail items.
4. **Re-rank**: MMR, novelty boosts, exposure quotas for new creators.
5. **Log impressions** to tell "not shown" from "shown and ignored".

Common mistakes

- Celebrating an offline win that comes only from recommending more popular items (test sets are popularity-biased too).
- Removing popularity entirely — it's real information; aim for calibration, not elimination.

TL;DR

Popular items get more data, so models over-recommend them and the loop reinforces itself. Measure, correct in training (logQ, IPS), explore, re-rank.

13.13 Real-world systems

Amazon's item-to-item CF is in §13.4.

Company	Publicly described approach	Ideas to remember
Netflix	Netflix Prize (2006–2009, $1M, won by BellKor's Pragmatic Chaos); personalised rows and ranking; artwork chosen per member with contextual bandits; recommender influences ~80% of hours streamed (2015)	Moved from star prediction to ranking on implicit signals; heavy A/B testing plus interleaving
YouTube	Deep candidate generation + ranking net predicting expected watch time (Covington et al., 2016); two-tower with streaming logQ (Yi et al., 2019); multi-task ranking (Zhao et al., 2019)	Two-stage funnel; watch time over clicks; "example age" feature for freshness
Spotify	Discover Weekly (2015) blends CF on listening and playlists, NLP on text about music, and audio models; bandits on home shelves; open-sourced Annoy and Voyager	Playlists as co-occurrence data; skips and saves as implicit feedback
Pinterest	PinSage (Ying et al., 2018): graph convolutional network over billions of pins and boards	The graph-embedding route

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Chapter TL;DR

- **Frame** as top-K ranking on a meaningful outcome; implicit feedback dominates; unobserved ≠ negative.
- **Content-based** covers new items; **collaborative** finds quality and surprise but needs history; hybrids win.
- **MF** ($\hat{r} = \mu + b_u + b_i + p_u \cdot q_i$; SGD or ALS; weighted ALS for implicit) is the strong baseline; its vectors are **embeddings**.
- **Two-tower** retrieval: items precomputed into an ANN index (IVF, HNSW, PQ); only the user tower runs per request; ship tower and index as one version.
- **Funnel**: retrieval (recall ceiling) → ranking (precision) → re-ranking (diversity, rules).
- **Evaluate** on a time-based split over the full catalogue against a popularity baseline, then A/B test. Cold start → content + exploration; counter popularity feedback loops.